Step-by-Step Guide:

Conducting a Systematic Literature Review (SLR)

Course:

Research Methodology

By:

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Topic: NLP for Domain-Specific Sentiment Analysis

Goal: To synthesize evidence on the performance of different NLP models for sentiment analysis when applied to specific domains like finance or healthcare.

Step 1: Define Your Research Question

A precise question is the foundation of your SLR. It guides every subsequent step. For a quantitative DS/AI topic, we can adapt the PICO framework.

- **Population/Problem:** Domain-specific text (e.g., financial news, patient reviews).
- **Intervention:** Modern NLP models (e.g., Transformer-based models like BERT).
- **Comparison:** Traditional ML models (e.g., SVM, Naive Bayes) or other baseline models.
- **Outcome:** Quantitative performance metrics (e.g., Accuracy, F1-Score, Precision, Recall).

Your Final Research Question:

In studies published from 2020 to 2025, how do Transformer-based NLP models (like BERT and its variants) compare against traditional machine learning models in terms of accuracy and F1-score for sentiment analysis on domain-specific datasets (e.g., finance, healthcare, legal)?

How to Use AI (ChatGPT): Open ChatGPT and use the following prompt to refine your question and create secondary questions to guide your search:

"Act as a research assistant. My primary SLR research question is: '[Paste your question here]'. Based on this, generate three secondary questions that explore the nuances of this topic, such as the impact of datasets, preprocessing techniques, and specific model architectures."

1. SRQ1
2. SRQ2
3. SRQ3

Step 2: Develop a Protocol – For Your SLR on NLP Domain Sentiment Analysis

Why a protocol?
A protocol is simply your full "game plan" before you start. It makes your process transparent for others (and for yourself in a few months!), and protects you from accidental bias or inconsistent searching.

A. Objectives

- **Main Objective:**
  To systematically identify, assess, and synthesize quantitative evidence about the performance of NLP models for domain-specific sentiment analysis.
- **Secondary Objectives:**
  - Compare transformer-based models (like BERT) with traditional/lighter-weight machine learning techniques for sentiment analysis.

- Identify which domains (e.g., finance, health, law) are most frequently studied.
- Summarize which datasets and metrics are most commonly used.

B. Eligibility Criteria

*Use these to **decide which studies to include/exclude** at every review/screening stage.*

- **Inclusion Criteria:**
  - Peer-reviewed journal papers and conferences
  - Published between January 1, 2020, and December 31, 2025
  - Written in English
  - Focused on domain-specific sentiment analysis (not "general sentiment")
  - Applies NLP models/algorithms AND reports quantitative performance (e.g., accuracy, F1-score)
- **Exclusion Criteria:**
  - Review papers, theoretical frameworks, qualitative-only studies
  - Papers on general-domain or unrelated NLP topics
  - Duplicate results, book chapters, editorials, patents
  - Studies lacking replicable experiments or metrics

C. Search Strategy

Databases to Search:

1. Google Scholar (https://scholar.google.com)
2. IEEE Xplore
3. ACM Digital Library
4. arXiv.org (for preprints)

Example Search String:
("Use in each database; slightly adapt as needed for search box syntax.")

("sentiment analysis" OR "opinion mining")
AND ("domain-specific" OR "finance" OR "health" OR "legal")
AND ("BERT" OR "Transformer" OR "LSTM" OR "deep learning" OR "NLP")
AND ("accuracy" OR "precision" OR "recall" OR "f1-score")
AND ("2020" OR "2021" OR "2022" OR "2023" OR "2024" OR "2025")

Tip: Save your exact search strings for later reporting.
Documentation: Record in a table: database, date searched, the exact query, number of results.

Bulk Search, Export, and Screening: The Workflow

Tool	Search Function	Bulk Export/Save	Imports to Zotero	Bulk PDF Download
Google Scholar	Yes	Using Zotero	Yes (via Connector)	Yes (if open)
IEEE/ACM	Yes	Yes (Citation Export	Yes	Yes (if open)
SpringerLink	Yes	Yes	Yes	Yes (if open)
arXiv	Yes	BibTeX/CSV	Yes	Yes (always free)

D. Study Selection Process

1. Title/Abstract Screening:

- First pass: Read titles/abstracts; exclude irrelevant studies.
- Track reason for exclusion in a simple sheet: "Not in domain," "No quantitative results," etc.

2. Full-Text Screening:

- Download and read PDFs of all remaining studies.
- Exclude any that don't meet all criteria upon closer reading, with reasons.

3. Duplicate Removal:

- Use a free reference manager (e.g., Zotero) to automatically detect and merge duplicates.

E. Data Extraction Plan

What data will you record for each paper?

- **General:** Author(s), year, title, publication venue
- **Study focus:** Domain (e.g., finance), dataset(s) used, model/method applied
- **Metrics:** Accuracy, F1-score, and any others reported
- **Model details:** Deep model (e.g., BERT) or traditional (e.g., SVM)
- **Baseline comparison:** Yes/No
- **Sample size** or data size
- Notes/comments

How to manage data?
Use a Google Sheet or Excel to create a table with the above column headers and fill it in for each included paper.

F. Quality Assessment Plan

- For each included study, score as follows:
    1. Clearly described methodology? (Y/N)
    2. Dataset transparency? (Y/N)
    3. Are all relevant metrics reported? (Y/N)
    4. Model reproducibility: enough details to replicate? (Y/N)
    5. Comparison to baseline or benchmarks? (Y/N)
- Make a column for each in your extraction sheet, and give 1 point for each "Y" (max 5 per paper).

G. Synthesis Plan

- **Quantitative synthesis:** Calculate and compare average metrics (accuracy, F1, etc.) for each model across domains.
- **Qualitative narrative:** Brief commentary on trends, gaps, and emerging topics.

H. PRISMA Reporting

- Track and record the number of papers at each phase:
    - Identified

    - After screening
    - After full-text review (with reasons for exclusion)
    - Final included studies
- Plan to produce a simple PRISMA flow diagram at the end.

Step 3: Conduct a Comprehensive Literature Search

This is where you gather all potentially relevant papers.

Recommended Free Tools:

- **Google Scholar:** The broadest starting point.
- **Zotero:** A free reference manager to collect, organize, and de-duplicate your findings.

Step-by-Step Process:

1. **Generate Search Strings with AI:** Use ChatGPT with this prompt:
    - *"Create a comprehensive search string for academic databases like IEEE Xplore and Google Scholar for my SLR question: '[Paste your question]'. Use Boolean operators (AND, OR, NOT) and include synonyms for key terms like 'sentiment analysis', 'NLP', 'BERT', and 'domain-specific'."*
    - **Example Output:** ("sentiment analysis" OR "opinion mining") AND ("natural language processing" OR "NLP") AND ("domain-specific" OR "finance" OR "medical" OR "legal") AND ("BERT" OR "Transformer" OR "LSTM") AND ("accuracy" OR "F1-score")
2. **Execute the Search:** Go to each database (Google Scholar, etc.) and run the search string.
3. **Export Results:** For each search, export the results as a .bib or .ris file. This is a standard format for citation data.
4. **Consolidate and De-duplicate:**
    - Create a free account for **Zotero** and download the desktop app.
    - Import all your .bib and .ris files into a new Zotero library.
    - In Zotero, click on the **"Duplicate Items"** collection in the left-hand pane. Zotero will automatically find duplicates, which you can then merge. This is a crucial step to clean your initial list

Step 4: Screen and Select Studies (The PRISMA Process)

This stage involves filtering the large pool of papers down to the ones that are truly relevant.

Recommended Free AI Tool:

- **Rayyan (rayyan.ai):** A web-based platform designed specifically for screening in SLRs. It is a no-code, collaborative, and AI-enhanced tool.

Step-by-Step Process:

1. **Export from Zotero:** Export your de-duplicated library from Zotero as a single .bib file.
2. **Import into Rayyan:** Create a free account on Rayyan and upload your .bib file to start a new review.
3. **Title and Abstract Screening:**
    - Rayyan will present you with the title and abstract of each paper.

- For each one, you will make a decision: **Include**, **Exclude**, or **Maybe**.
- **Use Rayyan's AI:** After you screen about 50 papers, click the **"Compute Ratings"** button. Rayyan's AI learns from your decisions and will give each remaining paper a 1- to 5-star rating, predicting how likely you are to include it.
- You can then sort the list by this rating to screen the most promising papers first, significantly speeding up the process.

4. **Full-Text Review (Eligibility):**
    - After the initial screening, you will have a smaller list of included papers.
    - For each of these, find the full PDF of the article.
    - Read the full paper and make a final decision based on your detailed eligibility criteria from Step 2. You will exclude many papers here because the full text might reveal they don't meet your criteria (e.g., they don't report the exact metrics you need). Track your reasons for exclusion (e.g., "Wrong Domain," "No Quantitative Results").

Step 5: Assess the Quality of the Studies

This step ensures the studies you include are methodologically sound.

Process:

1. **Create a Quality Checklist:** Develop a simple scoring system.
    - Was the research question clearly stated? (1 point)
    - Is the dataset used publicly available or well-described? (1 point)
    - Is the model architecture explained in detail? (1 point)
    - Is a comparison with a baseline model performed? (1 point)
    - Are the performance metrics clearly defined? (1 point)
    - **Total Score:** /5
2. **Apply to Included Studies:** For each paper in your final list, go through the checklist and assign it a quality score. You can use this score later to understand if higher-quality studies show different results from lower-quality ones.

Step 6: Data Extraction and Management

Here, you pull the specific pieces of information you need from each final paper.

Recommended Tool:

- **Google Sheets or Excel:** Simple, free, and effective.
- **Use Given 6 Sheet Sample**
    - **Create a new Excel workbook** named "SLR_NLP_Sentiment_Analysis_[YourName].xlsx"
    - **Create 6 sheets** with the exact names above
    - **Copy-paste each table** into the corresponding sheet
    - **Set up the formulas** as shown (the =SUM and =COUNTIF formulas will auto-calculate)
    - **Save the file** in a dedicated SLR folder on your computer

Process:

1. **Set up Your Spreadsheet:** Create a new Google Sheet. The columns should be the data points you defined in your protocol (Step 2): Paper_ID, Author, Year, Title, NLP_Model, Domain, Dataset, Accuracy, F1_Score, Quality_Score.
2. **Populate the Sheet:** For each included paper, meticulously fill in the data in the corresponding row. This is a manual but critical step. Be consistent with formatting.

Step 7: Data Synthesis

Now you analyze the extracted data to find patterns, trends, and answers to your research question. This is a form of **meta-analysis**.

Process (using Google Sheets):

1. **Descriptive Statistics:**
    - Use the SORT function to see which models, datasets, or domains are most common.
    - Use AVERAGE, MEDIAN, STDEV to calculate the central tendency and variance of performance metrics (e.g., average F1-score for BERT models).
2. **Comparative Analysis:**
    - Create **pivot tables** to compare performance across different categories. For example, set "NLP_Model" as rows, "Domain" as columns, and "Average of F1_Score" as the value. This will directly show you which models perform best in which domains.
3. **Visualize Findings:**
    - Create **bar charts** to compare the average accuracy of different model types (e.g., Transformers vs. Traditional ML).
    - Create **scatter plots** to see if there's a relationship between dataset size and model performance.

Step 8: Interpret the Findings

This is where you move from numbers to narrative. What does your data synthesis actually *mean*?

- **Summarize the Evidence:** "Our synthesis of 35 studies shows that Transformer-based models consistently outperform traditional ML models in the financial domain, with an average F1-score of 0.92 compared to 0.85."
- **Identify Gaps:** "Despite the focus on finance and healthcare, we found very few high-quality studies applying these models to the legal domain, representing a significant research gap."
- **Discuss Limitations:** Acknowledge limitations of your own review (e.g., "Our search was limited to English-language papers," or "Publication bias may favor studies with positive results").

Step 9: Report Your Findings

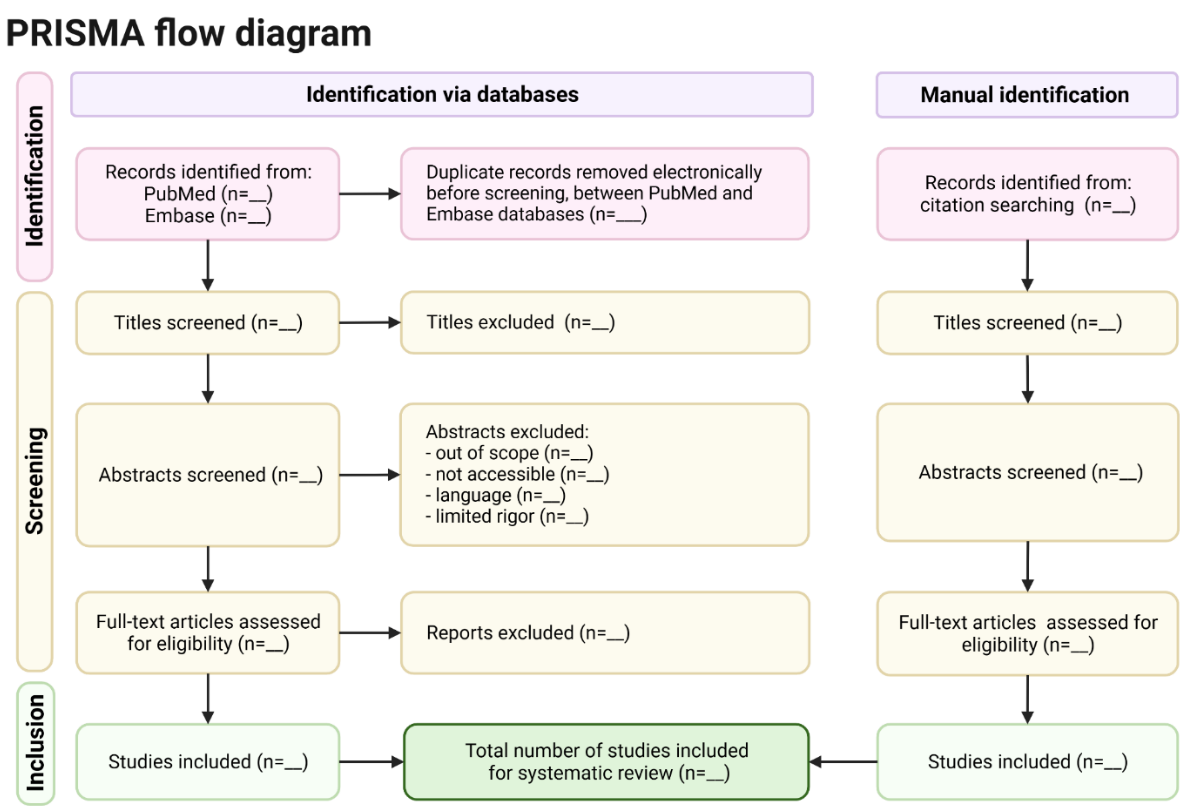
Your final report should be structured, clear, and follow the PRISMA guidelines.

Key Component: The PRISMA Flow Diagram This is a mandatory visual for any SLR. It shows the flow of information through the different phases of your review (identification, screening, eligibility, and inclusion).

How to Create it (No-Code): Use a free tool like **Canva** or **diagrams.net (formerly Draw.io)**. They have flowchart templates. Create four boxes to represent the main stages and use text to show the number of records at each phase:

1. **Identification:** Records identified from databases (n = [number from Step 3]).
2. **Screening:** Records after duplicates removed (n = ...) -> Records excluded (n = ...).

3. **Eligibility:** Full-text articles assessed for eligibility (n = ...) -> Full-text articles excluded, with reasons (n = ...).
4. **Included:** Studies included in final synthesis (n = [your final count]).



Step 10: Update Your Review

Knowledge evolves quickly in AI. An SLR can become outdated. Plan to re-run your search (Steps 3 and 4) in a year or two to find new studies and update your findings. Since you have a clear protocol, this process will be much faster the second time.